Since that first course in 2000, I began meditating for two hours every day, and each year I take a long meditation retreat of a month or two. It is not an escape from reality. It is getting in touch with reality. At least for two hours a day I actually observe reality as it is, while for the other twenty-two hours I get overwhelmed by emails and tweets and cute-puppy videos. Without the focus and clarity provided by this practice, I could not have written *Sapiens* or *Homo Deus*. At least for me, meditation never came into conflict with scientific research. Rather, it has been another valuable tool in the scientific toolkit, especially when trying to understand the human mind.

Digging from both ends

Science finds it hard to decipher the mysteries of the mind largely because we lack efficient tools. Many people, including many scientists, tend to confuse the mind with the brain, but they are really very different things. The brain is a material network of neurons, synapses and biochemicals. The mind is a flow of subjective experiences, such as pain, pleasure, anger and love. Biologists assume that the brain somehow produces the mind, and that biochemical reactions in billions of neurons somehow produce experiences such as pain and love. However, so far we have absolutely no explanation for how the mind emerges from the brain. How come when billions of neurons are firing electrical signals in a particular pattern, I feel pain, and when the neurons fire in a different pattern, I feel love? We haven't got a clue. Hence even if the mind indeed emerges from the brain, at least for now studying the mind is a different undertaking than studying the brain.

Brain research is progressing in leaps and bounds thanks to the help of microscopes, brain scanners and powerful computers. But we cannot see the mind through a microscope or a brain scanner. These devices enable us to detect biochemical and electrical activities in the brain, but do not give us any access to the subjective experiences associated with these activities. As of 2018, the only mind I can access directly is my own. If I want to know what other sentient beings are experiencing, I can do so only on the basis of second-hand reports, which naturally suffer from numerous distortions and limitations.

We could no doubt collect many second-hand reports from various people, and use statistics to identify recurring patterns. Such methods have enabled psychologists and brain scientists not only to gain a much better understanding of the mind, but also to improve and even save the lives of millions. However, it is hard to go beyond a certain point using only second-hand reports. In science, when you investigate a particular phenomenon, it is